

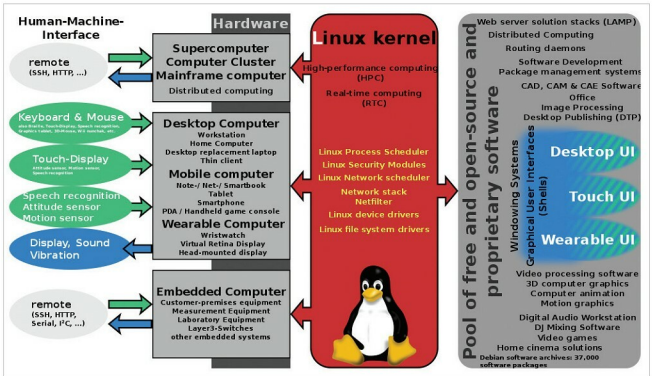


Figure 1: Inside a distribution

| Rank | Distribution               | Rank | Distribution                 |
|------|----------------------------|------|------------------------------|
| 1    | <a href="#">MX Linux</a>   | 11   | <a href="#">EndeavourOS</a>  |
| 2    | <a href="#">Manjaro</a>    | 12   | <a href="#">Zorin</a>        |
| 3    | <a href="#">Mint</a>       | 13   | <a href="#">openSUSE</a>     |
| 4    | <a href="#">Pop!_OS</a>    | 14   | <a href="#">deepin</a>       |
| 5    | <a href="#">Ubuntu</a>     | 15   | <a href="#">Arch</a>         |
| 6    | <a href="#">Debian</a>     | 16   | <a href="#">Ubuntu Kylin</a> |
| 7    | <a href="#">elementary</a> | 17   | <a href="#">Linuxfs</a>      |
| 8    | <a href="#">Fedora</a>     | 18   | <a href="#">antiX</a>        |
| 9    | <a href="#">Solus</a>      | 19   | <a href="#">CentOS</a>       |
| 10   | <a href="#">KDE neon</a>   | 20   | <a href="#">Puppy</a>        |

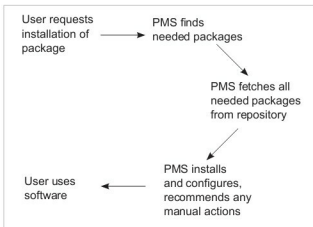
Figure 2: Top 20 distributions (Image source: [www.distrowatch.com](http://www.distrowatch.com))

Figure 3: How a package management system works

There are different types of package managers available for different distributions. Table 1 lists these.

This article focuses on Debian and Red Hat based package managers, as they are popular and have a large user base.

**Note:** Commands that start with \$ are meant for running from normal users. Commands that start with # are meant for running from administrator users. You need root privileges for package management.

A package manager or package management system is a collection of software tools that automate the process of installing, upgrading, configuring and removing computer programs from a computer's operating system in a consistent manner.

To give users more control over the kind of software that they allow to be installed on their systems (and sometimes due to legal or convenience reasons), software is often downloaded from a number of software repositories. All popular distributions maintain their own repositories. Here, the Internet is required to fetch the package from the repository.

## GUI based tools

You can use the synaptic package manager for managing packages in Linux distros. You can search, add and remove packages by only using the mouse. As this manager is very